

A computer-assisted all-subcritical Birkhoff global-section theorem at $\gamma = 459/1000$ for the planar circular restricted three-body problem

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Abstract

We prove a computer-assisted Birkhoff global-section theorem for the planar circular restricted three-body problem at the exact parameter $\gamma = 459/1000$, uniformly over all energies below the first critical level. After regularization, the proof is divided into two complementary energy regimes. Away from the critical level, an explicit exact symplectic momentum shear reduces dynamical convexity to strict convexity of the regularized double cover; the resulting positivity problem is certified by exact symbolic reduction, validated interval arithmetic, and an independent MPFR replay. In a thin neighborhood of the first critical value, where this global convexity mechanism is no longer uniform, we prove the required Conley–Zehnder lower bound directly from validated rotation estimates, short-period exclusion, finite return geometry, and a first-hit argument controlling the only region where the pointwise rotation estimate degenerates. Combining the two regimes shows that every contractible periodic orbit has disk-trivialized lower-semicontinuous Conley–Zehnder index at least three. Consequently, in each bounded regularized component, the standard retrograde orbit binds a rational open book whose disk-like pages are global surfaces of section. All theorem-critical computations are encoded by finite certificates and checked independently.

Keywords: restricted three-body problem; global surface of section; dynamical convexity; Conley–Zehnder index; computer-assisted proof; interval arithmetic.

1 Introduction

The planar circular restricted three-body problem (PCR3BP) is one of the classical testing grounds for the interaction between celestial mechanics, Hamiltonian dynamics, and modern symplectic geometry. In rotating coordinates it is a two-degree-of-freedom Hamiltonian system, but its regularized energy levels already display a rich mixture of periodic motion, bifurcation, transport, and collision geometry. A central idea going back to Birkhoff is that part of this complexity can be organized by a global surface of section: a two-dimensional surface whose interior is transverse to the flow and which is met repeatedly by every orbit in the relevant component. Such a section replaces a continuous three-dimensional flow by an area-preserving return map and therefore provides a natural framework for studying periodic orbits and global dynamics; see Birkhoff (1915).

Modern proofs of global sections in the PCR3BP pass through contact and symplectic geometry. Below the first critical value, collision regularization converts each bounded physical component into a compact contact-type energy hypersurface; see Albers et al. (2012a) and the systematic account in Frauenfelder and van Koert (2018). Global sections in important PCR3BP regimes were established by holomorphic-curve methods in Albers et al. (2012b); the broader finite-energy-foliation framework is developed in Hofer et al. (2003). A particularly

effective route is dynamical convexity: in the convention used here, every contractible periodic Reeb orbit must have lower-semicontinuous disk-trivialized Conley–Zehnder index at least three. Strict convexity of an S^3 -type energy hypersurface in $\mathbb{R}^4$ implies this index bound through the work of Hofer, Wysocki and Zehnder (Hofer et al., 1998); for the rational-open-book conclusion on $\mathbb{R}P^3$, we use Hryniewicz and Salomão (2016).

The index-theoretic viewpoint is already visible in the rotating Kepler limit, where the Conley–Zehnder indices of all subcritical periodic orbits can be determined explicitly and dynamical convexity follows on the universal cover (Albers et al., 2013). Validated computation becomes useful when the corresponding convexity or index inequalities in the PCR3BP are explicit but not accessible by hand over the entire parameter domain. In the PCR3BP, Jouné and van Koert (2025) developed computational symplectic methods for symmetric periodic orbits and Conley–Zehnder indices. More recently, Liu and Salomão (2025) proved finite-energy-foliation results and, in particular, an all-subcritical Birkhoff conclusion for mass ratios sufficiently close to the equal-mass case. The neighborhood in that theorem is existential for the comparison needed here. The present paper therefore makes a different fixed-parameter statement: it treats the exact auxiliary parameter

$$\gamma = \frac{459}{1000},$$

and proves the Birkhoff global-section conclusion for every subcritical energy at that parameter. No interval of mass ratios is claimed.

The main technical point is that one geometric mechanism does not remain uniformly effective all the way to the first critical level. We therefore divide the proof at the exact threshold

$$\Delta_0 = \frac{3}{50000}, \quad \Delta = c - c_{L_1}(\gamma) > 0.$$

For $\Delta \geq \Delta_0$, an explicit exact symplectic momentum shear converts the regularized double cover into a strictly convex hypersurface. Exact elimination reduces the tangent-Hessian condition to a finite positivity problem in two base variables, which is certified with validated arithmetic and independently replayed. For $0 < \Delta < \Delta_0$, strict Euclidean convexity is replaced by a direct rotation-index argument. The near-critical proof combines a pointwise rotation inequality, a short-period exclusion, a finite return graph, a long-orbit occupation contradiction, and a first-hit decomposition of the only block where the baseline pointwise rotation estimate is not uniform.

The proof is computer-assisted but finite in the strong sense needed for auditing. Search and adaptive subdivision are used only to generate finite proof objects. The theorem-critical claims are then accepted by exact arithmetic, validated interval inequalities, complete finite coverage, and independent fail-closed replay/checkers. The purpose of the computational layer is therefore not to sample trajectories, but to certify a finite list of mathematical inequalities and covering obligations from which the theorem follows.

Figure 2 gives a compact view of how the two regimes meet in the final theorem. The rest of the paper is organized as follows. Section 2 fixes the Hamiltonian, parameter, regularization, time-change, and index conventions. Section 3 proves the deep subcritical branch by strict convexity after an exact shear. Section 4 develops the near-critical rotation and return estimates, while Section 5 closes the exceptional first-hit block. Section 6 assembles the two regimes into dynamical convexity and the global-section conclusion. Section 7 explains the computer-assisted proof architecture and reproducibility, and Section 8 records the relation to previous work and the precise scope of the theorem.

1.1 Exact parameter and main theorem

We fix

$$\gamma = \frac{459}{1000}.$$

Define

$$D(\gamma) = 1 - 2\gamma + \gamma^2 + 2\gamma^3 - \gamma^4, \quad m_s(\gamma) = \frac{\gamma^3(\gamma^2 - 3\gamma + 3)}{D(\gamma)}.$$

For $0 < \gamma \leq 1/2$, $m_s(\gamma)$ is the smaller-mass representative and is strictly increasing. At $\gamma_* = 459/1000$,

$$m_s(\gamma_*) = \frac{177321681763299}{441699674239000} = 0.4014530508060817\dots$$

The labeling used internally in the regularized Hamiltonian is the complementary one,

$$\mu_* = 1 - m_s(\gamma_*) = \frac{264377992475701}{441699674239000} = 0.5985469491939183\dots$$

Thus the apparently larger value $\mu_* > 1/2$ is only a choice of primary labeling; after exchanging the primaries, the conventional representative is $m_s(\gamma_*) \leq 1/2$. The first critical Jacobi value is also rational in this parameterization:

$$c_{L_1}(\gamma_*) = \frac{388389323017759836541758}{195098602222838720229121} = 1.990733498818957869\dots$$

We use the Jacobi convention $\Sigma_c = H^{-1}(-c)$ and define

$$\Delta = c - c_{L_1}(\gamma).$$

Thus $\Delta > 0$ is the subcritical side.

Table 1: Exact parameter and convention data used throughout the proof.

Item	Exact convention/value	Publication role
Exact parameter coordinate	$\gamma = 459/1000$	Fixed throughout; no mass interval is claimed.
Mass convention	$m_s(\gamma_*) \approx 0.401453 \leq 1/2$, while $\mu_* = 1 - m_s(\gamma_*)$ is the complementary label	Makes the primary-exchange convention explicit and prevents interpreting γ itself as the standard mass ratio.
Energy convention	$\Sigma_c = H^{-1}(-c)$, $\Delta = c - c_{L_1}(\gamma)$	$\Delta > 0$ is the subcritical side.
Deep/near split	$\Delta_0 = 3/50000$	Strict-convexity propagation for $\Delta \geq \Delta_0$; rotation/return proof for $0 < \Delta < \Delta_0$.
Index convention	Lower-semicontinuous, disk-trivialized μ_{CZ} for contractible orbits	Dynamical convexity means $\mu_{CZ}^{\text{disk}} \geq 3$.
LC/time semantics	Two focus-specific Levi-Civita charts with positive time factors; reduced frame/trivialization transported with the cocycle	No global equal-clock identification is used.

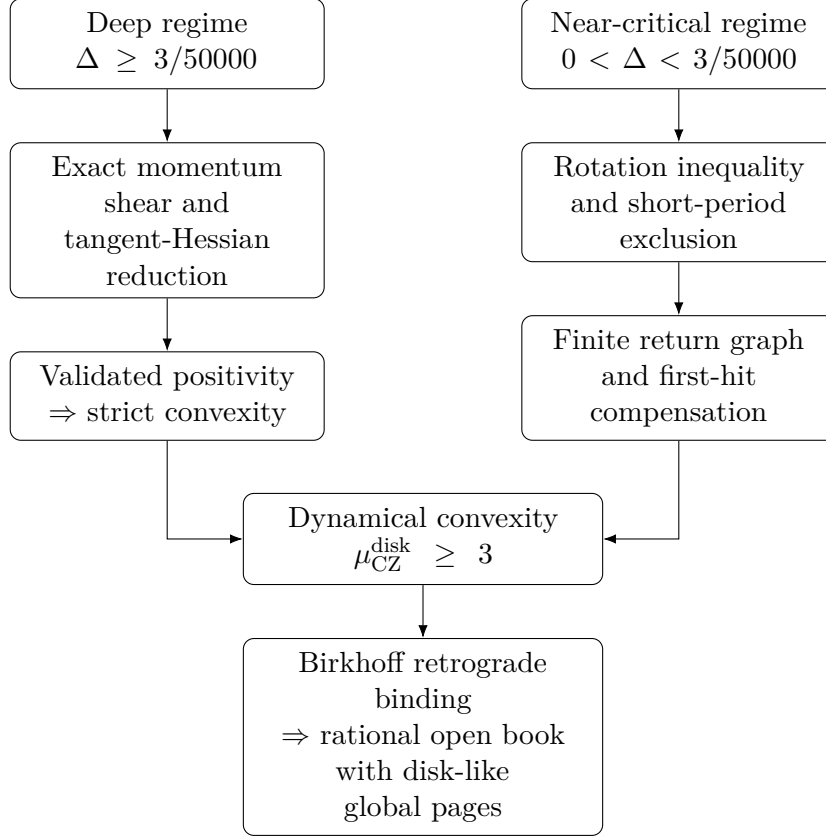


Figure 2: Logical architecture of the proof. The two energy regimes use different mechanisms but meet at the same index conclusion, after which the contact-topological open-book theorem gives the global surfaces of section.

2.2 Why an exact parameter coordinate is used instead of a decimal mass ratio

The exact coordinate γ is introduced to keep the collinear critical data rational rather than solving numerically for the first critical point. With

$$D(\gamma) = 1 - 2\gamma + \gamma^2 + 2\gamma^3 - \gamma^4, \quad m_s(\gamma) = \frac{\gamma^3(\gamma^2 - 3\gamma + 3)}{D(\gamma)},$$

one has, for $0 < \gamma \leq 1/2$,

$$m'_s(\gamma) = \frac{\gamma^2(1 - \gamma)^2(9 - 6\gamma + 5\gamma^2 + 2\gamma^3 - \gamma^4)}{D(\gamma)^2} > 0.$$

Hence γ is a one-to-one exact rational coordinate for the smaller physical mass ratio, up to exchange of the primaries. At the parameter used here, $m_s(459/1000) = 1 - \mu_*$ as recorded above. In particular, γ is not being used as a synonym for the conventional PCR3BP mass ratio μ .

In this parameterization the first collinear point is represented by

$$x_{L_1} = 1 - m_s(\gamma) - \gamma,$$

and the first critical Jacobi value is obtained exactly from

$$c_{L_1}(\gamma) = \frac{x_{L_1}^2}{2} + \frac{1 - m_s(\gamma)}{1 - \gamma} + \frac{m_s(\gamma)}{\gamma}.$$

At $\gamma = 459/1000$ this gives the rational value stated in Section 1.1.

2.3 Energy and regularization conventions

We use the Jacobi convention

$$\Sigma_c = H^{-1}(-c), \quad \Delta = c - c_{L_1}(\gamma_*).$$

Thus increasing c moves downward in Hamiltonian energy, and $\Delta > 0$ is the subcritical side. At the first critical point L_1 , $H(L_1) = -c_{L_1}$.

Collision regularization compactifies each bounded subcritical physical component. In the convention used here, its regularization double cover is diffeomorphic to S^3 , and the physical quotient is diffeomorphic to $\mathbb{R}P^3$. The exact momentum shear used in Section 3 is compatible with the deck involution, so it descends to the quotient.

The near-critical argument uses two focus-specific Levi–Civita charts, one for each bounded component. Their Hamiltonian time changes are positive. Reduced linearized cocycles are compared after transporting the contact-plane frame/trivialization through these changes. Consequently, the Conley–Zehnder and Robbin–Salamon data used below are invariant under the transported conjugacy; no equality of the two time parameters is assumed. For the geometric regularization framework see Moser (1970) and Frauenfelder and van Koert (2018); for the index formalism see Robbin and Salamon (1993).

3 Fixed-energy convexity anchor and deep-energy propagation

3.1 Exact global symplectic shear

At the exact parameter $\gamma = 459/1000$, the regularized Hamiltonian admits the exact global momentum shear generated in physical coordinates by

$$G(q) = -\frac{53}{250}q_1q_2.$$

In elliptic-hyperbolic regularizing coordinates this is equivalent to

$$g(x) = \sinh(2x_1) \sin(2x_2) = 16q_1q_2, \quad \lambda = -\frac{53}{4000}.$$

Because the momentum translation is by an exact one-form, it is an exact symplectomorphism, and the generator is compatible with the deck involution.

3.2 Exact tangent-Hessian reduction

After the shear the Hamiltonian remains in the magnetic-mechanical class

$$H(y, x) = \frac{1}{2}|y + F(x)|^2 + V(x).$$

We use the generalized tangent-Hessian criterion of Liu and Salomão (Theorem 9.1 in arXiv:2506.17867v2) (Liu and Salomão, 2025). Exact symbolic elimination of the fiber angle reduces the positivity problem to

$$L(s, t) = A_0(s, t) - \sqrt{R_1(s, t)} - \sqrt{R_2(s, t)} > 0,$$

where $s = \cosh x_1$, $t = \cos x_2$, and A_0, R_1, R_2 are exact rational bivariate polynomials.

The fixed-energy certificate has 13,655 final leaves: 11,902 PASS, 1,753 OUTSIDE, and no PENDING leaf. The smallest certified lower bound is

$$8.331573711073198 \times 10^{-9} > 0.$$

An independent replay regenerated the exact polynomial data, rebuilt GCC and Clang computations, reconstructed the final ledger byte-for-byte, independently audited the tiling, and replayed all 13,655 leaves with MPFR-192 directed rounding.

3.3 Rebased energy-ray propagation

The base threshold is $\Delta_0 = 3/50000$. At this level the finite strict-convexity certificate closes, and exact energy-ray monotonicity propagates positivity for every $\Delta \geq \Delta_0$. The independent replay checks all 15,364 global terminal leaves with MPFR-192 and zero failures, including the 4,207 residual leaves inherited from the fixed-energy computation. Therefore all levels $\Delta \geq 3/50000$ are dynamically convex through strict convexity.

Proposition 3.1 (Deep subcritical regime). *For every $\Delta \geq 3/50000$, each bounded regularized component is dynamically convex. More precisely, after the exact global momentum shear described above, the regularized S^3 lift is strictly convex, and hence every contractible periodic Reeb orbit has disk-trivialized Conley–Zehnder index at least three.*

4 Near-critical rotation and finite return geometry

We now restrict to $0 < \Delta < 3/50000$. The argument is by contradiction. Assume that a contractible periodic orbit has disk-trivialized $\mu_{CZ} \leq 2$. We first convert this index assumption into an upper bound on the total admissible rotation. A validated lower bound on the instantaneous rotation then produces an occupation-time budget. Short periodic trajectories are excluded separately. Any remaining low-index orbit must therefore be long, and the finite return geometry forces it to revisit a residual region so many times that the occupation budget is violated. Section 5 treats the only local block where the baseline pointwise rotation estimate is not uniform by pairing each residence with a preceding first-hit entry segment.

4.1 Segmentwise rotation estimate

Let Θ denote the angular variation of the transported reduced linearized flow in the chosen contact-plane trivialization. The validated quantity $\beta(w)$ is a pointwise lower bound for its instantaneous rotation, so that

$$\Theta(b) - \Theta(a) \geq \int_a^b \beta(w(t)) dt.$$

The rotation-index convention used here gives

$$\Theta(T) > 2\pi \implies \mu_{CZ}^{\text{disk}} \geq 3.$$

On the finite near-critical chart partition, the certificate verifies all 32,661 parent obligations for the target $\beta \geq 1/100$ on the region where the baseline pointwise bound is intended to hold.

The same framework uses stronger regions where $\beta \geq 2/5$ and $\beta \geq 7/200$. Let t_{40} and t_{35} denote the certified occupation times in those two stronger regions, and let t_R be the remaining time to which the baseline bound $\beta \geq 1/100$ applies. Thus

$$T = t_{40} + t_{35} + t_R.$$

The exact occupation identity used by the exact-arithmetic checker is

$$\frac{2}{5}t_{40} + \frac{7}{200}t_{35} + \frac{1}{100}t_R = \frac{1}{100}T + \frac{39}{100}t_{40} + \frac{1}{40}t_{35}.$$

Consequently the segmentwise estimate gives

$$\Theta(T) \geq \frac{1}{100}T + \frac{39}{100}t_{40} + \frac{1}{40}t_{35}.$$

If a contractible periodic orbit had $\mu_{\text{CZ}}^{\text{disk}} \leq 2$, the rotation-index implication forces $\Theta(T) \leq 2\pi$. Multiplication by 200 therefore gives the exact low-index occupation budget

$$2T + 78t_{40} + 5t_{35} \leq 400\pi.$$

The convention audit separately verifies the Robbin–Salamon/Conley–Zehnder comparison and the complementary contribution used in this rotation estimate. In particular, the argument never identifies the two Levi–Civita time variables: the reduced frame and cocycle are transported through the positive time changes.

4.2 Current-neck short-period exclusion

The current-neck region is controlled by validated Yorke/Wirtinger-type period estimates, exact energy support, and displacement inequalities. The certified conclusion is

$$T_{\widehat{H}} > 3$$

for every nonconstant periodic orbit represented in the finite current-neck cover. The independent replay reconstructs exactly the 1,222 boxes: 1,212 close by the unweighted Yorke test, nine of the ten residual boxes close by the fixed-diagonal weighted Yorke test, and the single remaining index closes by the independent scalar-displacement certificate. Thus the accounting is 1,222/1,222, with no missing or extra cover element.

4.3 Finite return graph and long regime

The residual return geometry is encoded by a finite θ -ordered graph. An independent checker validates all 30/30 theorem-critical half-phase transitions. The resulting conservative hard graph has 34 edges, no self edge, no θ -order violation, is acyclic, and has maximum hard-edge path length 5. The only consequence used below is therefore transparent: a completed residual cluster contains at most seven residual visits.

An independent propagation replay then checks the validated obligations that make this counting argument quantitative: 2,392/2,392 strong-window obligations, 2,362/2,362 outer safe half-obligations, 10/10 hard-sink half-obligations, and 2,372/2,372 detachment half-obligations pass. The exact residence estimate is

$$T_{\text{res}} < \frac{145\pi}{106496} < \frac{1}{230},$$

while the occupation argument gives $t_R \geq 219/100$. Hence a hypothetical long low-index orbit must make at least 504 residual visits. Since the return graph allows at most seven visits per completed cluster, at least 72 clusters are required. The exact cluster arithmetic contradicts the available long-regime occupation budget. This closes the hypothetical low-index regime $T > 400\pi/7$ without enumerating periodic orbits.

5 The exceptional first-hit block

The previous rotation argument uses the baseline estimate $\beta \geq 1/100$ pointwise. The last local obstruction is a small region where that pointwise statement is not true, so it must be handled without weakening the global rotation budget. The region arises from an exact 16-cell seed cover. Fourteen cells satisfy the pointwise theorem $\beta \geq 1/100$. Two cells, denoted *s04_d0* and *s06_d0*, do not. They are never relabeled as pointwise $\beta \geq 1/100$ cells; instead, their union is controlled by an integrated first-hit argument.

5.1 One-way bad block and corridor

Let $B = s04_d0 \cup s06_d0$. On B , and on a larger surrounding corridor, validated inequalities give

$$\dot{\theta} < 0, \quad \dot{A} < 0,$$

and the two ϕ faces point strictly outward. Hence local recirculation is impossible. The entry structure used in the proof is summarized schematically in Figure 3.

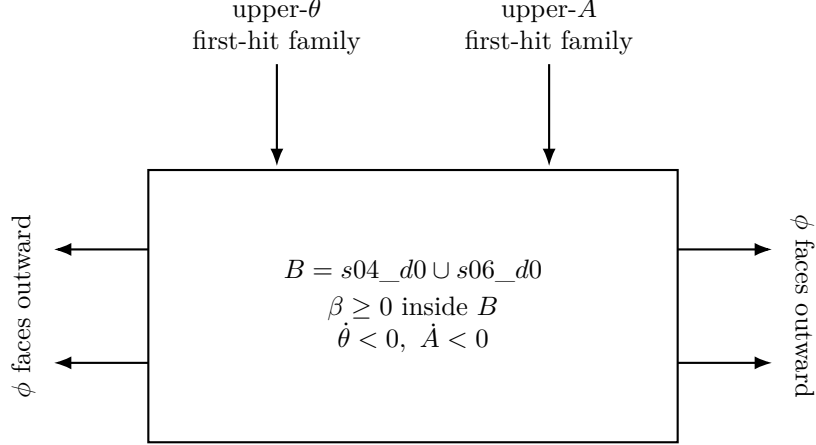


Figure 3: Schematic first-hit geometry of the exceptional block B ; the drawing is not a coordinate projection and is not to scale. The validated face signs exclude forward entry through the two ϕ faces. Hence every maximal residence in B is paired with exactly one preceding upper- θ or upper- A first-hit segment, which supplies the integrated rotation compensation used in Lemma 5.1.

5.2 Theta-entry geometry

On the complete theta-entry shell, $\dot{A} < 0$. All possible negative β is confined to a narrow exact sublayer N , where

$$\beta \geq -\frac{3}{200}$$

and $\dot{A}$ is sufficiently negative to give

$$\beta - \frac{3}{20}\dot{A} > 0.$$

Inside B itself, $\beta \geq 0$ on both exceptional cells.

5.3 Upper-theta entry family

The complete upper-theta entry face has the exact natural partition

$$32 \times 32 \times 3 = 3072$$

roots. The finite genealogy closes all roots and yields

$$J_{001} > 0.005708670873993645.$$

The generic following- B deficit is smaller than 0.0049211744370937422, hence the paired upper-theta margin is

$$> 0.0007874964368999028 > 0.$$

5.4 Upper-A entry family

On the high- A corridor, $\beta \geq 1/20$, and every upper- A -reaching first-hit block spends more than $1/40$ of the relevant time there. Relative to the $\beta = 1/100$ target this yields

$$J_{001}(\text{entry}) > \left(\frac{1}{25}\right) \left(\frac{1}{40}\right) = \frac{1}{1000}.$$

The following B residence satisfies $T_{B,\text{upperA}} < 17/200$, so the deficit relative to $1/100$ is $< 17/20000$. Therefore

$$\frac{1}{1000} - \frac{17}{20000} = \frac{3}{20000} > 0.$$

Lemma 5.1 (First-hit decomposition of B). *Decompose an orbit into maximal connected residence intervals in B . Every forward entry into B occurs through exactly one of two families: the upper- θ face or the upper- A face. Pair each maximal residence with its immediately preceding first-hit entry block. These pairs are disjoint in time and no entry block is reused.*

Proof. The outward ϕ -face signs exclude forward ϕ entry. The strict θ and A monotonicities leave only the upper- θ and upper- A faces as forward entry faces. Maximal connected residences are pairwise disjoint by definition. The immediately preceding first-hit block terminates at the corresponding entry event and cannot simultaneously precede a distinct maximal residence. $\square$

Corollary 5.2. *Each entry-plus- B pair satisfies the effective $\beta = 1/100$ rotation budget. Hence the two exceptional seed cells are discharged by integrated rotation, and the complete 16-cell cover is closed.*

Proposition 5.3 (Near-critical regime). *For every $0 < \Delta < 3/50000$, each bounded regularized component is dynamically convex. In particular, the segmentwise rotation estimate, short-period exclusion, finite return graph, long-orbit occupation contradiction, and first-hit control of the exceptional block exclude every contractible periodic orbit with disk-trivialized Conley–Zehnder index at most two.*

6 Proof of the main theorem

Proposition 3.1 covers $\Delta \geq 3/50000$, while Proposition 5.3 covers $0 < \Delta < 3/50000$. These intervals form the complete subcritical ray $\Delta > 0$. In the deep branch, strict convexity gives dynamical convexity by (Hofer et al., 1998). In the near-critical branch, assume a contractible periodic orbit had disk-trivialized $\mu_{\text{CZ}} \leq 2$. The two-component Levi–Civita semantics, segmentwise rotation inequality, current-neck exclusion, finite return graph, long-regime contradiction, and complete first-hit discharge reduce every possible itinerary to one of the validated closed cases, contradicting the low-index assumption. Thus every contractible orbit has $\mu_{\text{CZ}}^{\text{disk}} \geq 3$.

The exact shear preserves disk-trivialized CZ under transported capping-disk trivializations. For the regularization double cover $S^3 \rightarrow \mathbb{R}P^3$, a contractible capping disk lifts, and the pulled-back contact planes and linearized flows give the same disk-trivialized index upstairs and downstairs.

The standard Birkhoff retrograde orbit on a subcritical regularized component has the 2-unknot and rational self-linking $-1/2$ topology used in the literature; we use the discussion and Theorem 1.5 of arXiv:2506.17867v2 (Liu and Salomão, 2025). Hryniewicz and Salomão prove the rational-open-book/global-section implication for a p -unknotted orbit with self-linking $-1/p$ in a dynamically convex Reeb flow on a lens space of order p ; in arXiv:1505.02713v3 this is Corollary 1.8 (Hryniewicz and Salomão, 2016). The authors later issued a correction to a

broader lens-space claim in the introduction; the results of the paper are unchanged, and the correction explicitly notes that the $L(2, 1) = \mathbb{R}P^3$ case used here remains valid (Hryniewicz and Salomão, 2019). Taking $p = 2$ gives the final statement of Theorem 1.1.

7 Computer-assisted proof architecture and reproducibility

The computational part is organized so that search is separated from theorem acceptance. Adaptive subdivision, numerical scouting, and exploratory floating-point integrations may suggest a finite cover or a useful decomposition, but none of these objects is accepted as proof by itself. A theorem-critical claim is accepted only after it has been converted into a finite proof object whose obligations are checked with exact arithmetic or validated interval arithmetic.

There are four recurring layers. First, an exact symbolic layer fixes the parameter conventions and reduces the relevant inequalities to explicit algebraic expressions. Second, a generator constructs a finite cover together with local obligations such as positivity, exclusion, transition, or residence-time bounds. Third, coverage is checked independently: every admissible point must belong either to a certified PASS element or to a mathematically justified OUTSIDE element, with no unresolved gap. Fourth, an independent replay/checker reevaluates the finite obligations using directed rounding; the deepest replays use MPFR multiple-precision arithmetic, whose correct-rounding implementation is described by Fousse et al. (2007). For the general interval-arithmetic framework see Moore (1966).

This distinction is important. The theorem does not rely on the behavior of a search heuristic, on visual plots, or on an unverified numerical trajectory. It relies on finitely many explicit inequalities and covering statements. The search program may be replaced without changing the proof, provided that the same finite obligations are produced and accepted by the checker.

Table 2 summarizes the theorem-facing evidence. Detailed package identifiers, full SHA-256 values, compiler information, and the finer Claim–Evidence genealogy belong to the archival reproducibility release rather than to the logical narrative of the proof.

Table 2: Traceability map for theorem-critical computer-assisted components.

Proof component	Finite mathematical evidence	Independent acceptance
Exact parameter and fixed-energy anchor	Exact rational parameter reconstruction, symbolic tangent-Hessian reduction, and a 13,655-leaf cover	Exact reconstruction plus independent MPFR-192 replay and tiling audit
Deep-energy propagation	15,364 terminal obligations at the rebased threshold and monotone energy-ray propagation	15,364/15,364 independent MPFR-192 acceptance
Rotation and index bridge	Finite near-critical chart cover, transported reduced-frame convention, and exact occupation identity	Independent semantic/index audit plus finite replay of the certified chart obligations
Short- and long-orbit control	1,222 current-neck boxes, 30 return transitions, acyclic hard graph, residence and propagation bounds	Independent short-period, transition-graph, and propagation checkers
Exceptional first-hit block	Complete 16-cell cover, 3072 upper- θ entry roots, upper- A entry bounds, and paired integrated margins	Independent coverage and entry-assembly replay
Final assembly	Dependency map connecting the deep and near-critical propositions to Theorem 1.1	Claim–Evidence and release-manifest audit

8 Relation to previous work and scope

Joung and van Koert developed validated computational methods for symmetric periodic orbits and Conley–Zehnder indices in the PCR3BP (Joung and van Koert, 2025). Liu and Salomão prove, among other results, Birkhoff’s retrograde-orbit conclusion for mass ratios sufficiently close to $1/2$ and all subcritical energies; the relevant all-subcritical statement appears as Theorem 1.16 in arXiv:2506.17867v2 (Liu and Salomão, 2025). The neighborhood in that result is existential for the comparison made here. We therefore do not claim that the parameter $\gamma = 459/1000$ lies outside their theorem.

This paper proves a fixed-parameter all-subcritical theorem at $\gamma = 459/1000$. It does not prove an interval of mass ratios, optimality of the shear, uniqueness or nondegeneracy of all periodic orbits, or that every retrograde-looking numerical orbit is a valid rational-open-book binding. Earlier exploratory computations, broad orbit searches, and failed global rotation thresholds are provenance, not premises of the final proof.

9 Reproducibility and availability

The archival release will contain the dependency map, Claim–Evidence Matrix, exact parameter and environment manifests, theorem-critical proof objects, independent checkers, a quick-audit route, a complete replay route, and SHA-256 manifests. Persistent repository and DOI identifiers will be inserted only after the release assets are finalized.

Statements and Declarations

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Competing interests. The author declares no competing interests.

Author contributions. The author conceived the study, developed the mathematical and computational methodology, implemented the software, performed the formal analysis and validated computations, curated the proof objects and research data, and wrote and revised the manuscript.

Data availability. The theorem-critical finite certificates, manifests and audit objects will be deposited in the final public reproducibility release. [Persistent identifier to be inserted after deposit.]

Code availability. The proof-critical source, replay scripts and independent checkers will be released with the archival reproducibility package. [Persistent identifier to be inserted after deposit.]

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